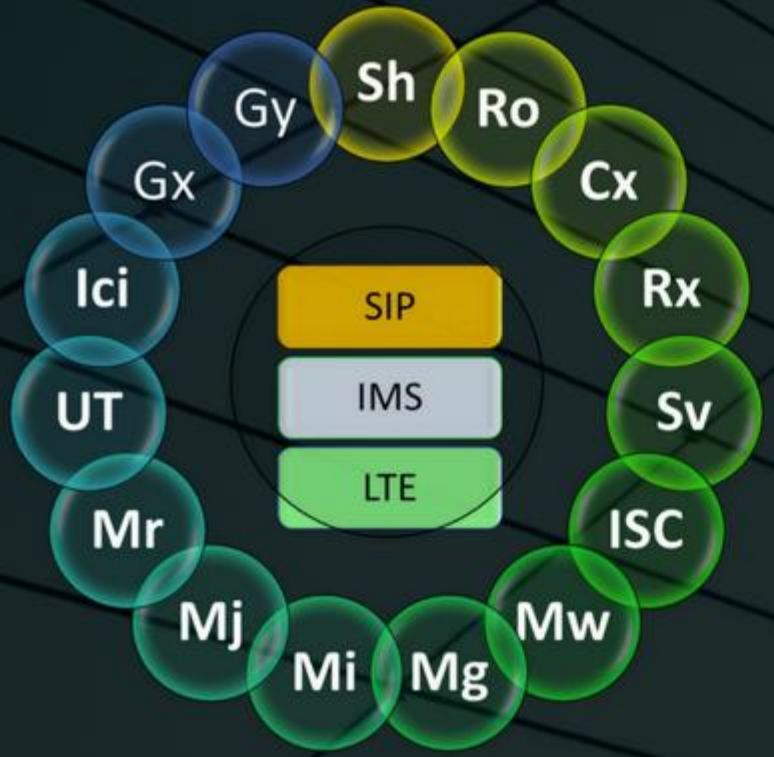
VOLTE IMS



- Links
- Interfaces
- Protocol Stack

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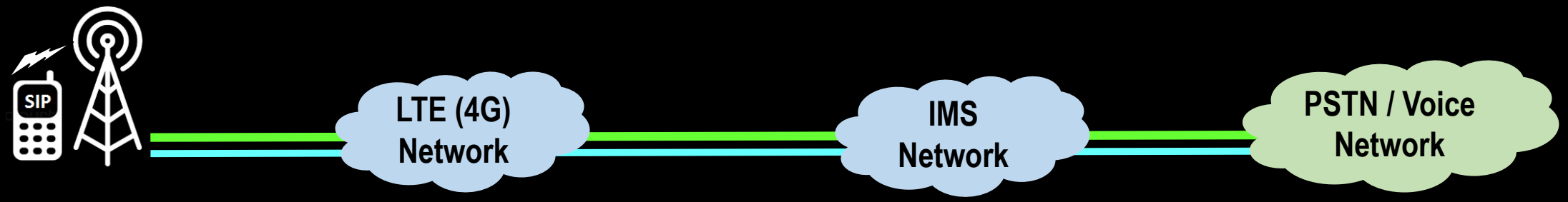
LTE / VoLTE IMS Protocols Interfaces



LTE User Plane	LTE Control Plane	IMS User Plane	IMS Control Plane
Ex : S1 , S5, SGi	*Ex : S1 , S10 , S11 , S6*	*Ex : Gm , SIP , Izi*	*Ex : Sh , Ro , Cx , Rx , Sv , ISC , Mw , Mg , Mi*

UDP TCP SCTP

IP Layer

MAC / Data Link Layer

Physical / Ethernet

LTE Protocol Stack

IMS Protocol Stack

RTP

Diameter

SIP / SDP

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SCTP Vs UDP

SCTP (**Stream Control Transmission Protocol**)

- Requires Strong / Robust Media connectivity
- Provides Super Reliable Communication
- Connection-oriented protocol similar to TCP
- Provides facilities such as multi-streaming and multi-homing
- Better performance and redundancy
- S1AP , X2AP , Diameter (S6 , Gx , Gy)
- Best Suited for Control Plane & Close Reliable Network such as Billing , Charging & Authentications

UDP (**User Datagram Protocol**)

- Suited for Non-Reliable Media
- Does not guarantee that Data in the order on receive
- Minimal message-oriented transport layer protocol
- Enables two hosts to connect and send short messages to one another
- User Plane : S1U , GTPv1U (S5 , S8)
- Control Plane : GTPv2C (S5 , S8)
- Best Suited for Long Distance & Open Networks such as Roaming etc..

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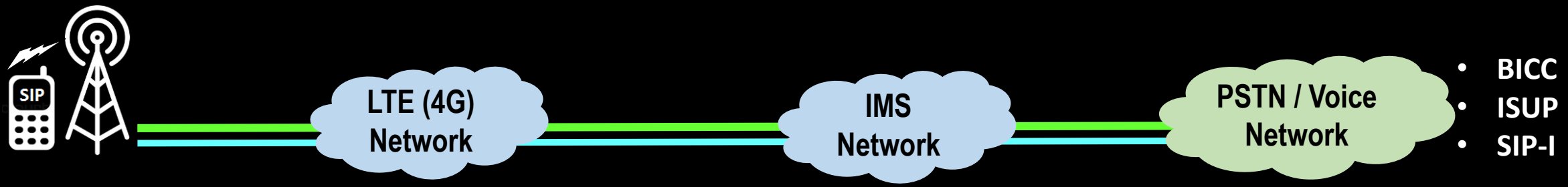
LTE / IMS Protocols Interfaces



LTE User Plane	LTE Control Plane
• S1U • S5/S8 • SGi	• S1MME • S5/S8 • S10 • S11 • S6a • Gx • Gy • X2 • SGs
GTPv1U , S1AP , SGi	GTPv2C , Diameter , AP (X2AP , SGSAP , S1AP)

IMS User Plane	IMS Control Plane
• Gm • SIP • Izi	• Sh • Ro • Cx • Rx • Sv • ISC • Mw • Mg • Mi • Mj • Mr / Mr‘ • UT • Ici
RTP , RTCP	SIP & SDP , HTTP , Diameter , Megaco

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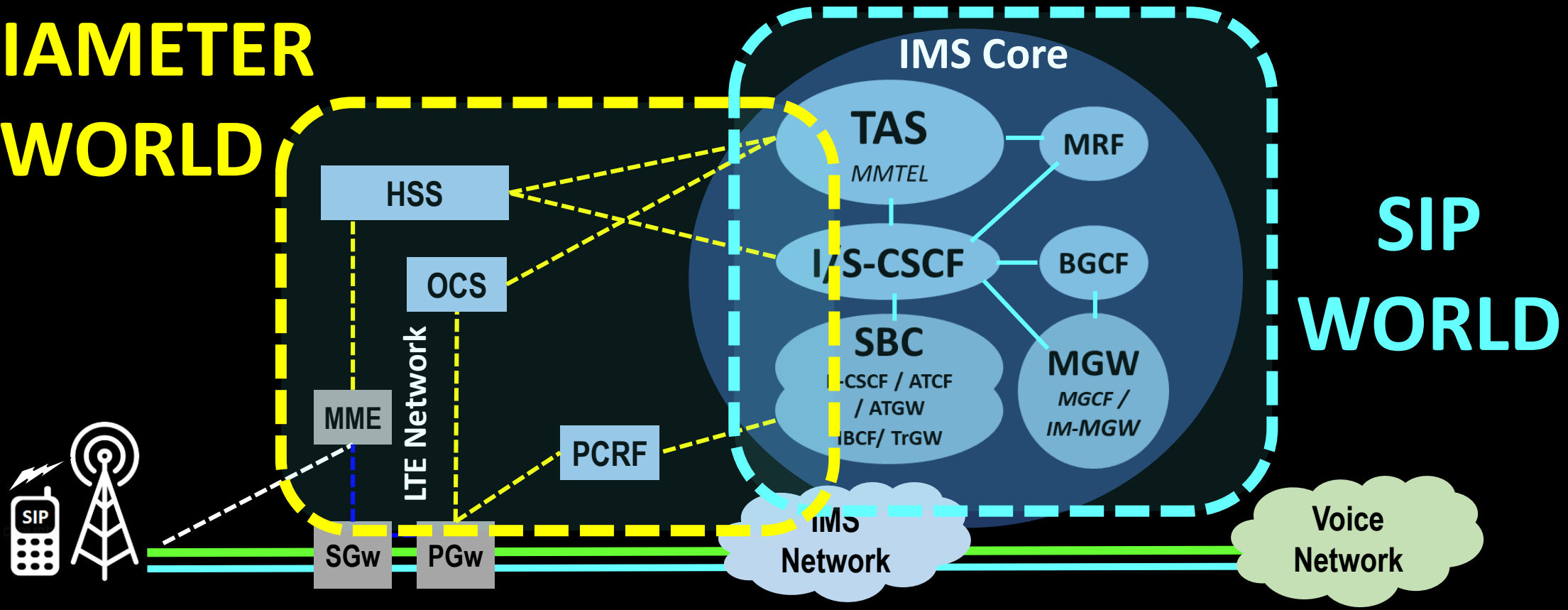
VoLTE IMS Connectivity Overview



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SIP & Diameter Protocols

SIP (Session Initiated Protocol)

- Used for Controlling Voice / Multimedia Calls & Uses RTP / RTCP for media plane
- SIP manages set-up/establishing, tying together, and tear-down/ terminating of multimedia communications
- 5 Functions of SIP :- a) *User Location b) User Availability c) User Capabilities d) Session Setup e) Session Management*

Diameter

- Evolution of RADIUS (AAA) Protocol - Authentication, authorization, and accounting (AAA) services
- Reliable & Secure (Runs on TCP / SCTP)
- Supports Failover mechanism to maintain redundancy
- Support for Proxy & Relay Agents
- Support for Capability Negotiation via Mandatory / Optional AVPs
- Provides Dynamic discovery of Peers

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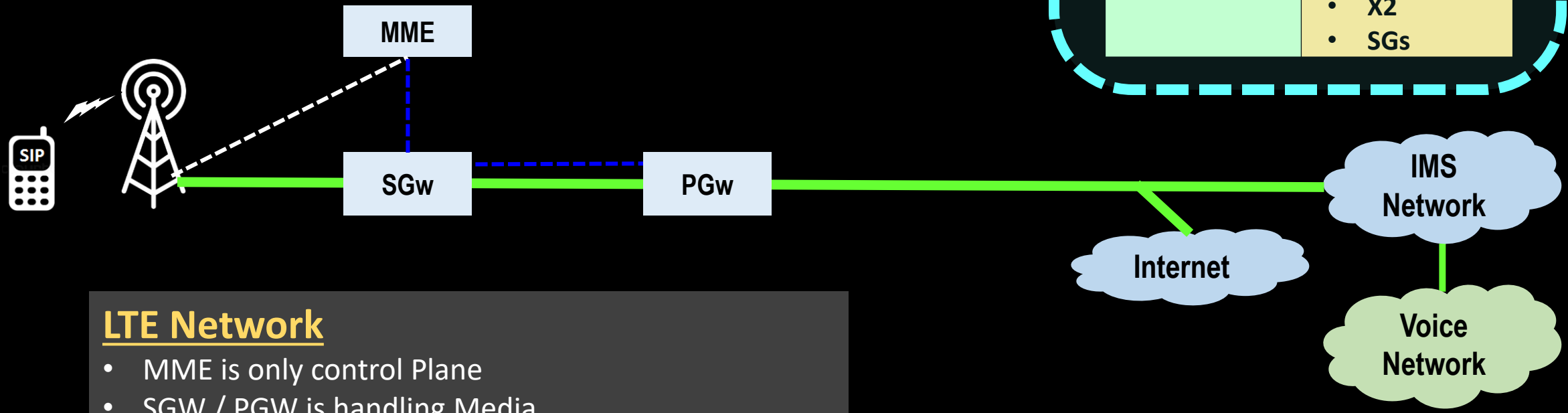
LTE Network – Protocols & Links

LTE User Plane	LTE Control Plane
• S1U • S5/S8 • SGi	• S1MME • S5/S8 • S10 • S11 • S6a • Gx • Gy • X2 • SGs



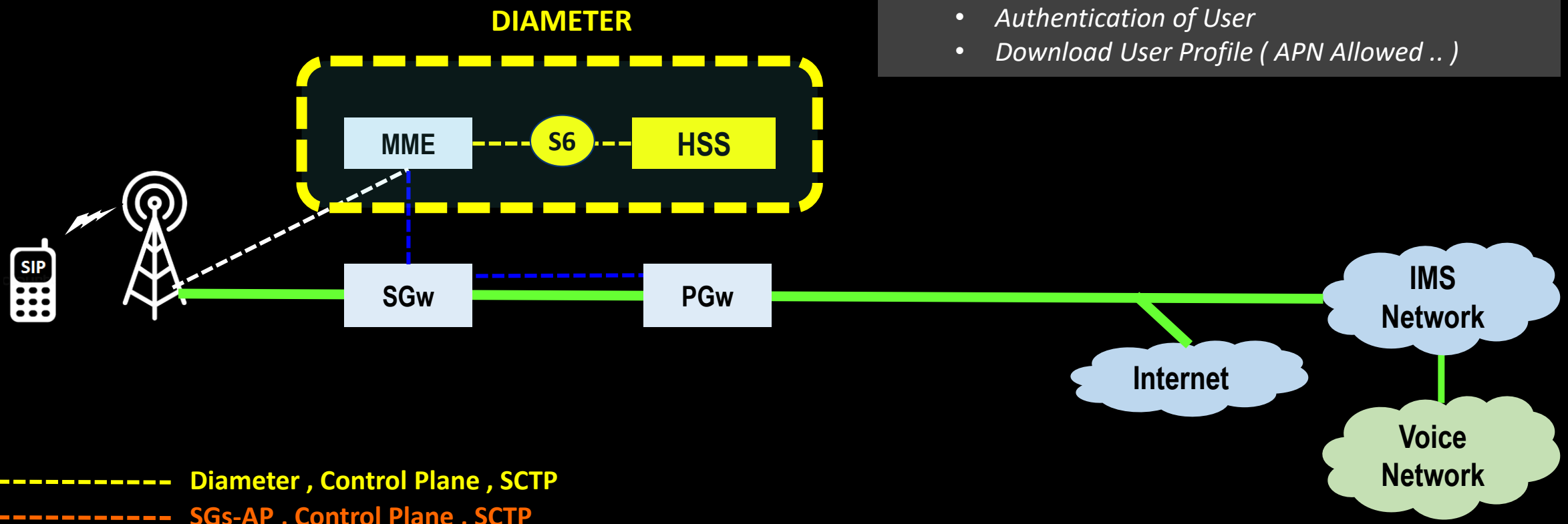
LTE Network

- MME is only control Plane
- SGW / PGW is handling Media

S6a – LTE Protocols & Links

S6 Interface (MME to HSS)

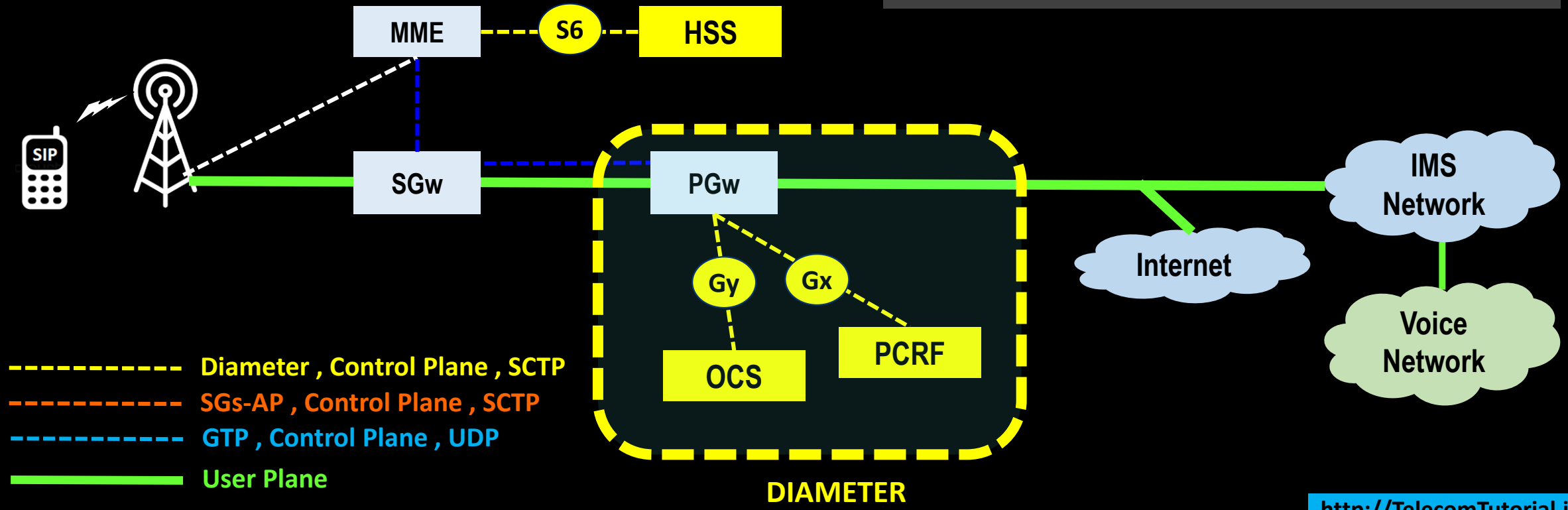
- Diameter Link connects MME to HSS
- Used for :-
  - *Authentication of User*
  - *Download User Profile (APN Allowed ..)*



Gx & Gy – LTE Protocols & Links

Gx & Gy Links

- Gx : Diameter Link connects PGW to PCRF
  - *Used for QOS , Traffic Prioritization*
- Gy : Diameter Link connects PGW to OCS
  - *Used for Balance Reservation , Internet Quota Allocation & Management , Wallet for Services*



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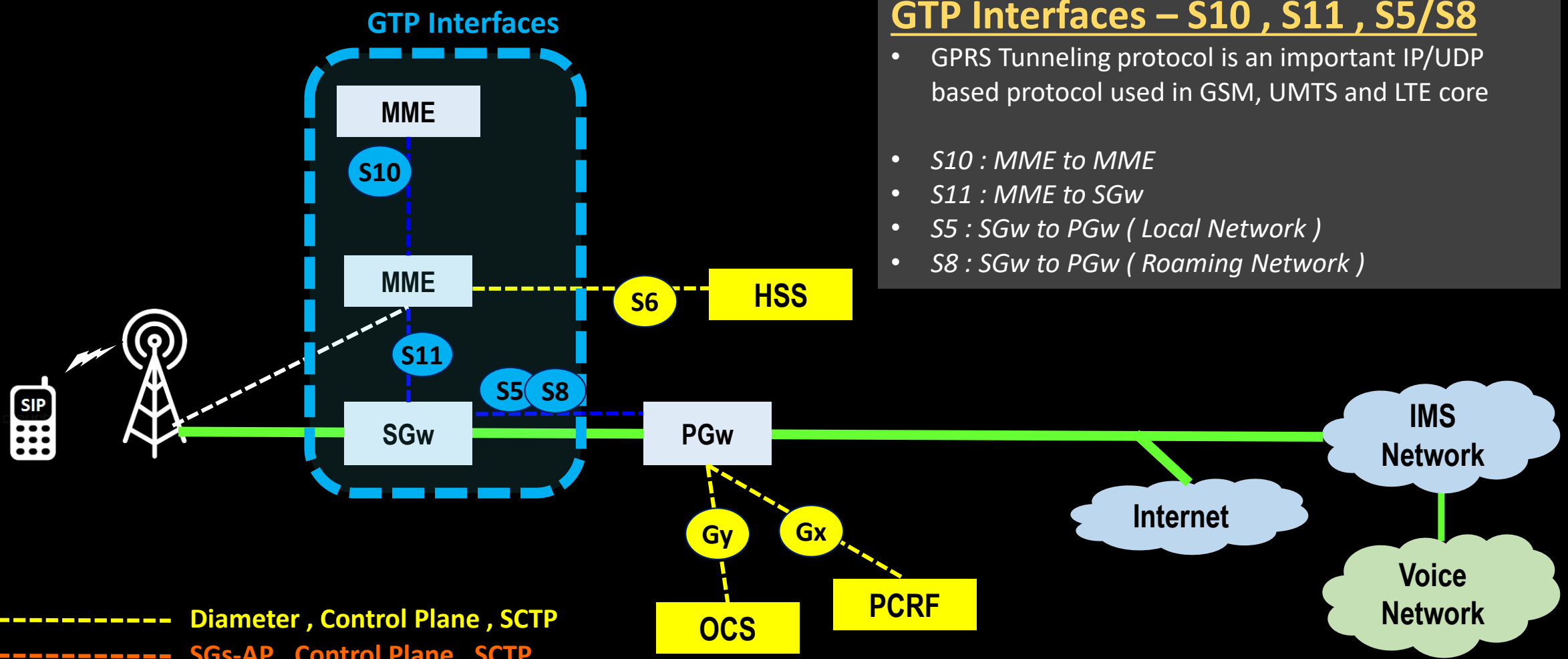
GTP Interfaces – LTE Protocols & Links



GTP Interfaces – S10 , S11 , S5/S8

- GPRS Tunneling protocol is an important IP/UDP based protocol used in GSM, UMTS and LTE core

- *S10 : MME to MME*
- *S11 : MME to SGw*
- *S5 : SGw to PGw (Local Network)*
- *S8 : SGw to PGw (Roaming Network)*

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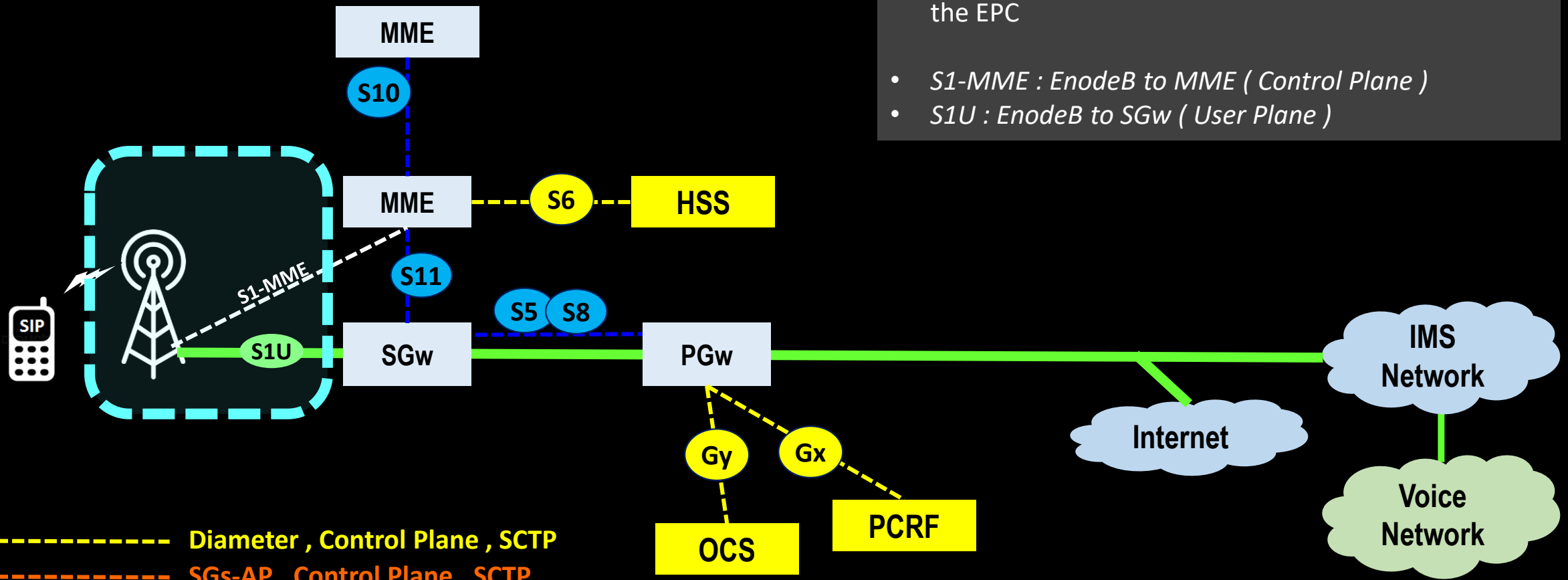
S1MME / S1U – LTE Protocols & Links

S1 Interface

- The S1 interface in LTE is used between EnodeBs and the EPC

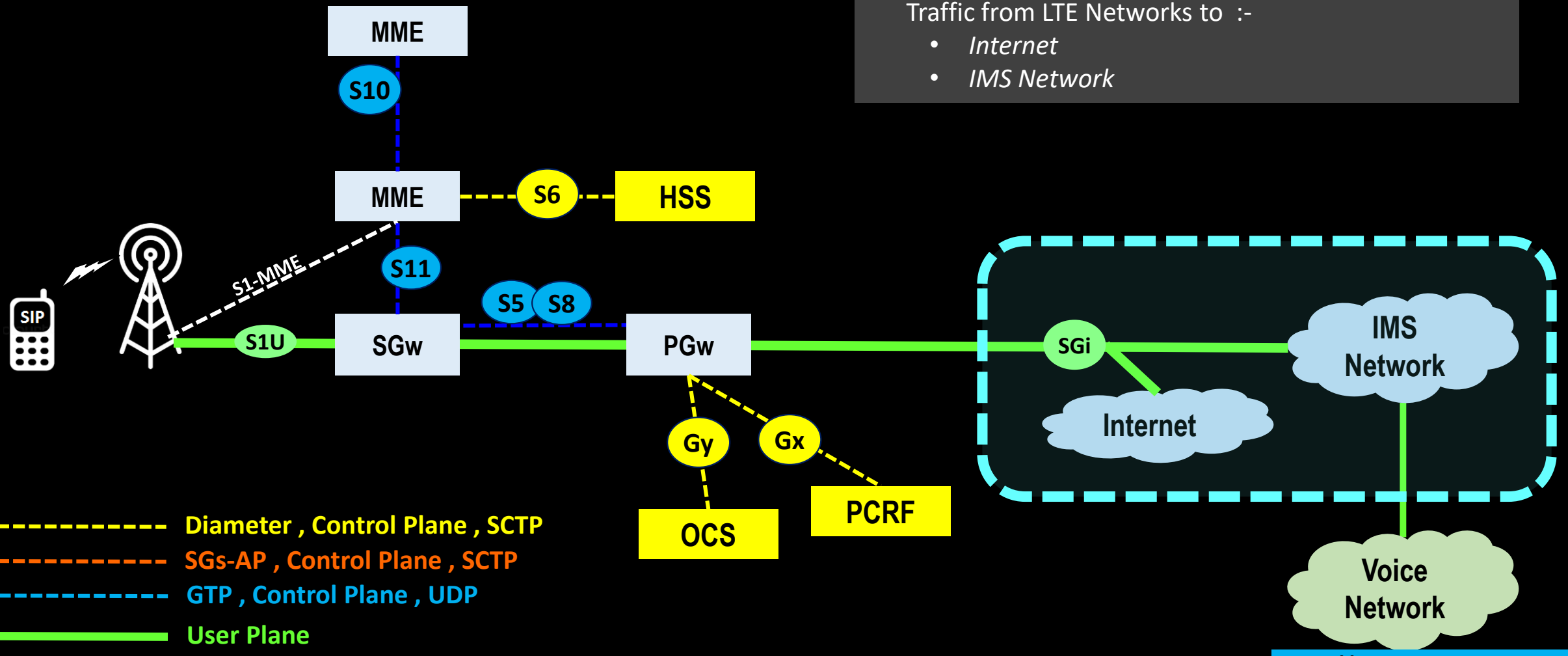
- *S1-MME : EnodeB to MME (Control Plane)*
- *S1U : EnodeB to SGw (User Plane)*



SGi Link – LTE Protocols & Links

SGi Interface

- SGi link is IP / TCP Interface Used as Exit path for Traffic from LTE Networks to :-
  - *Internet*
  - *IMS Network*



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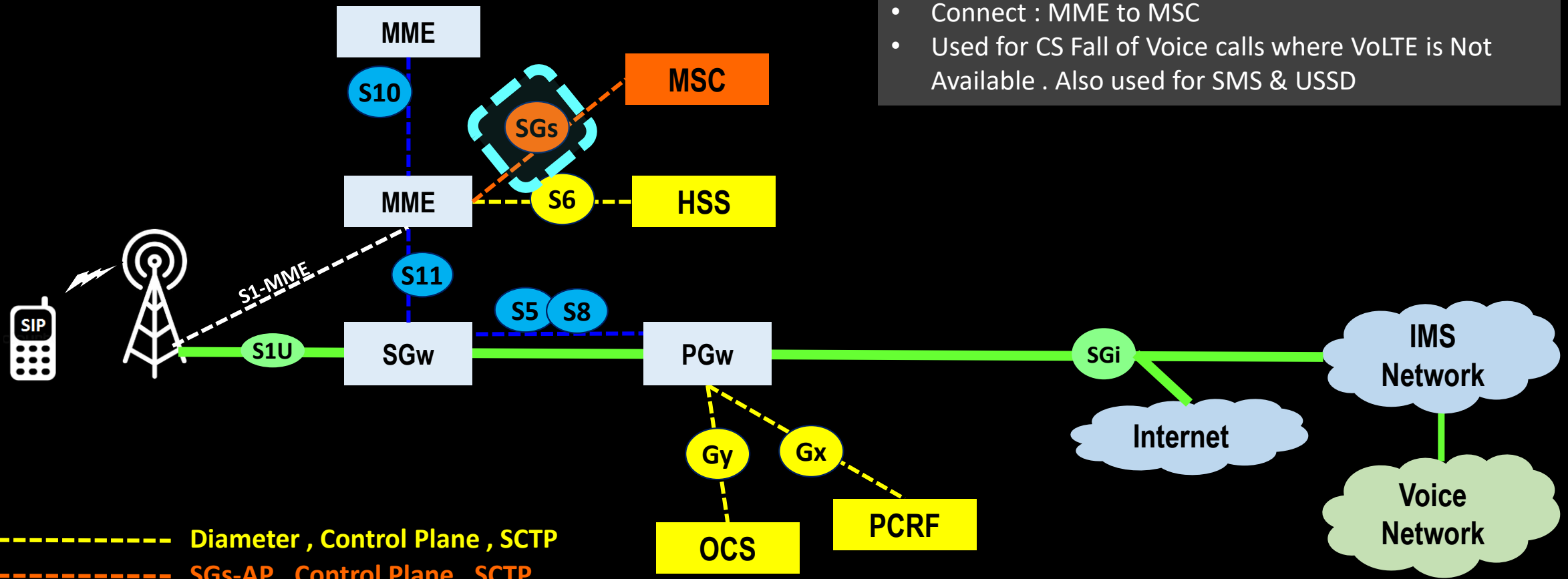
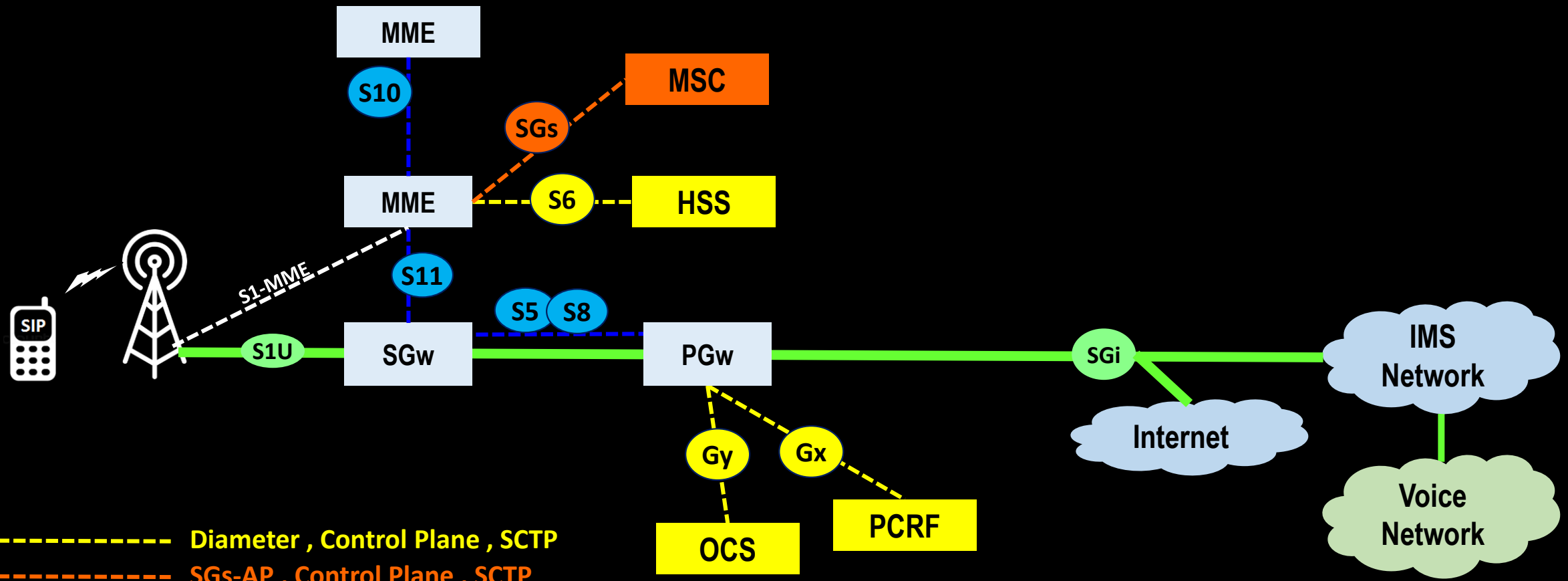
SGs Link – LTE Protocols & Links

SGs Interface

- Runs on SGsAP Protocol over SCTP
- Connect : MME to MSC
- Used for CS Fall of Voice calls where VoLTE is Not Available . Also used for SMS & USSD



Complete LTE Network – Protocols & Links



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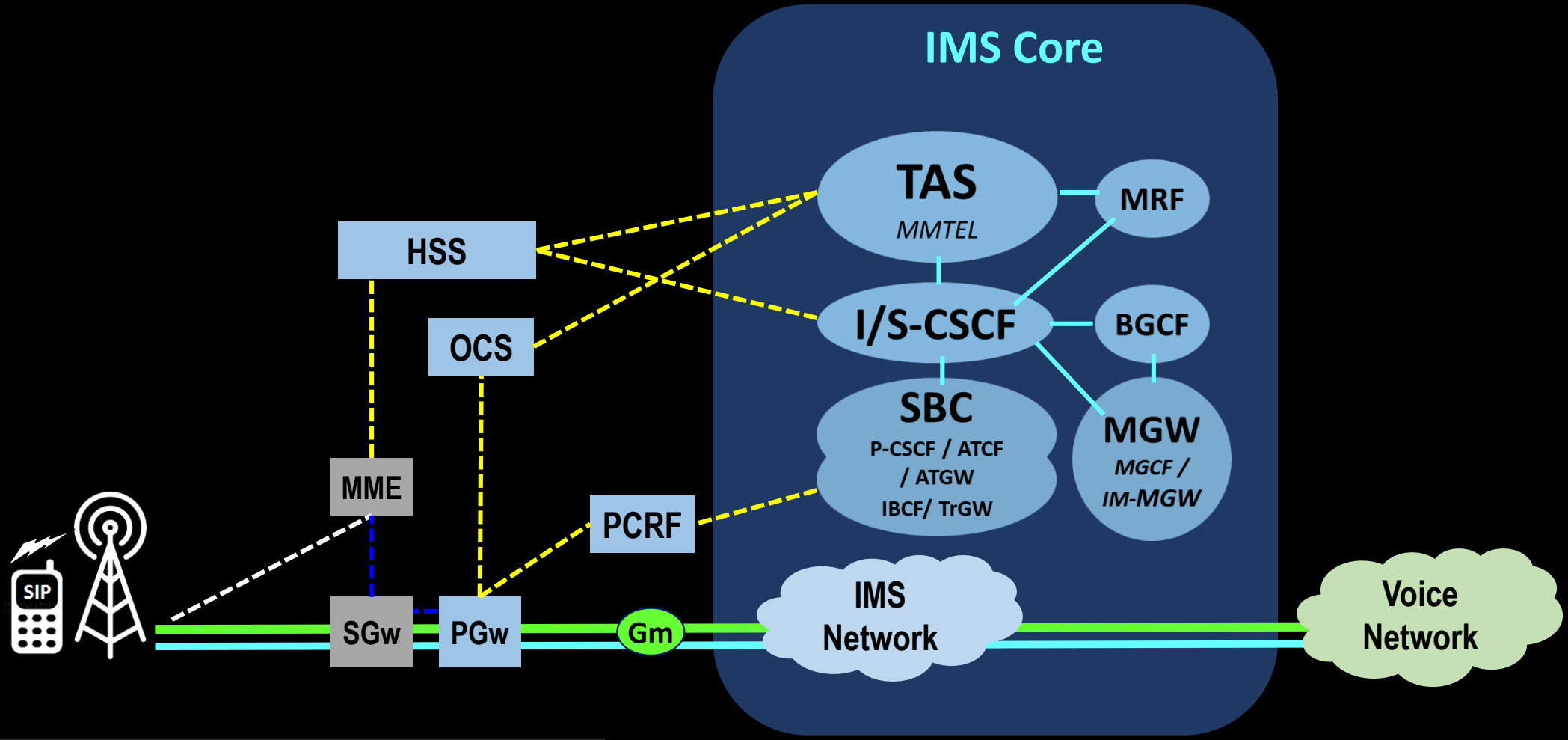
VoLTE IMS Connectivity Overview



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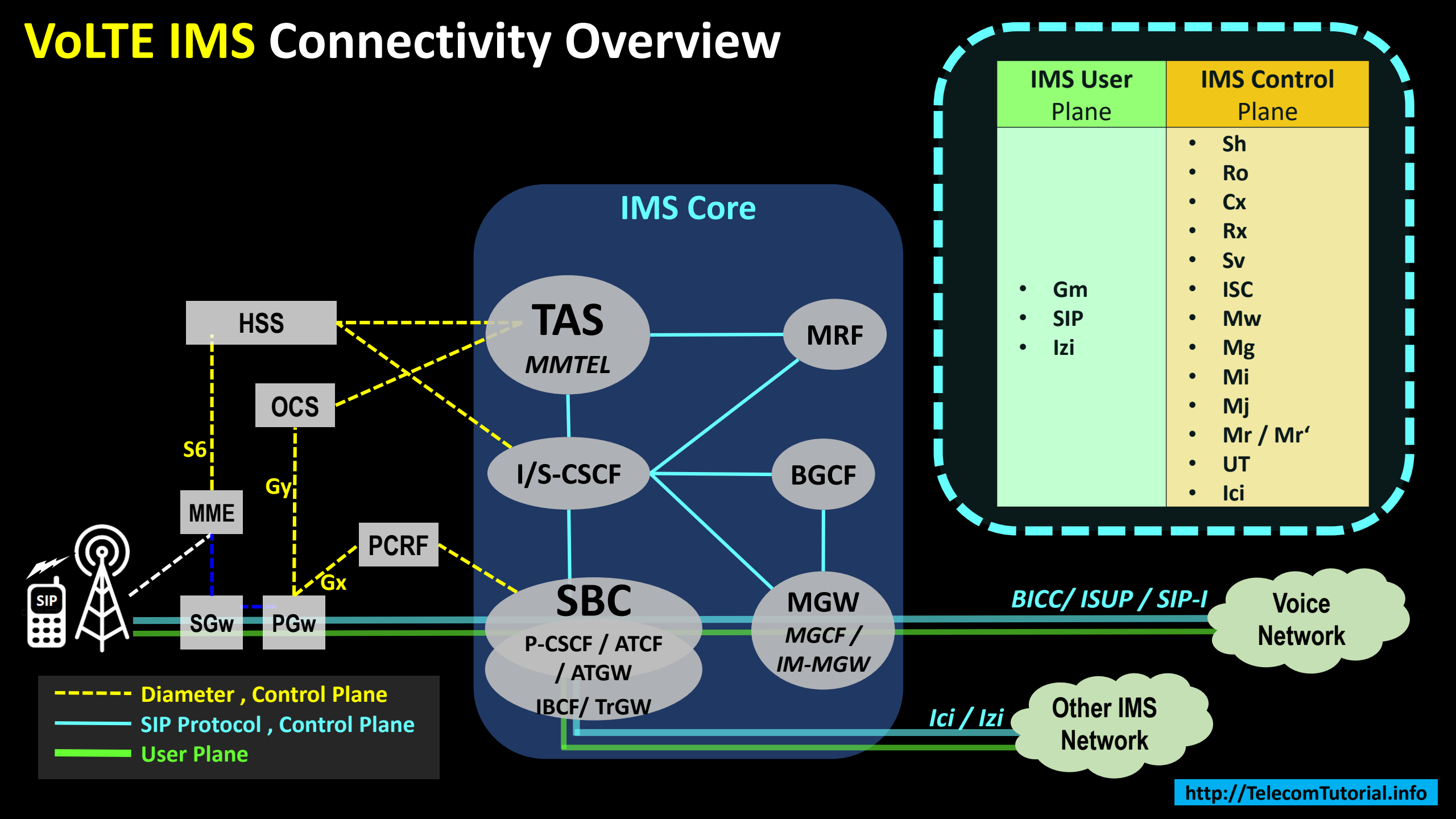
VoLTE IMS Connectivity Overview



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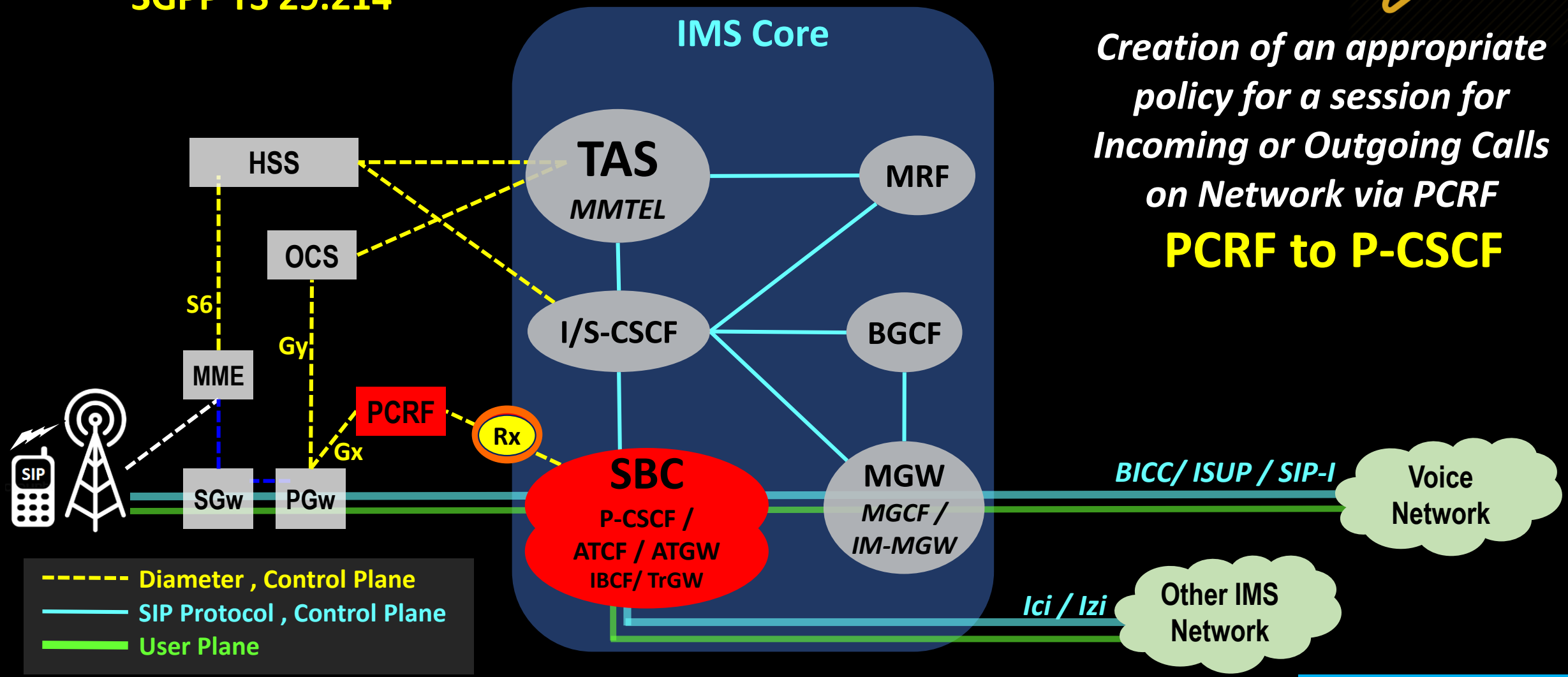
Rx DIAMETER

3GPP TS 29.214

Creation of an appropriate policy for a session for Incoming or Outgoing Calls on Network via PCRF

PCRF to P-CSCF



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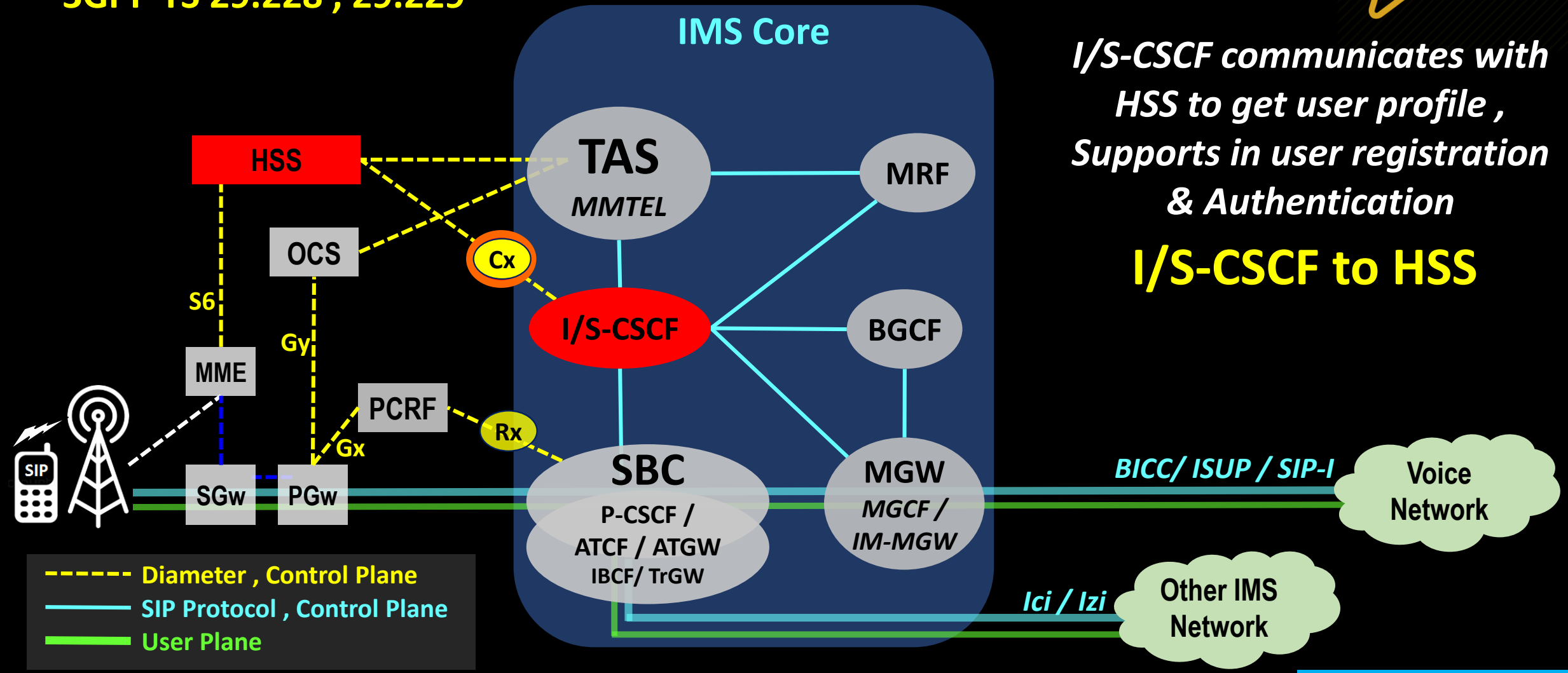
Cx DIAMETER

3GPP TS 29.228 , 29.229



WHY? Cx

I/S-CSCF communicates with HSS to get user profile , Supports in user registration & Authentication

I/S-CSCF to HSS

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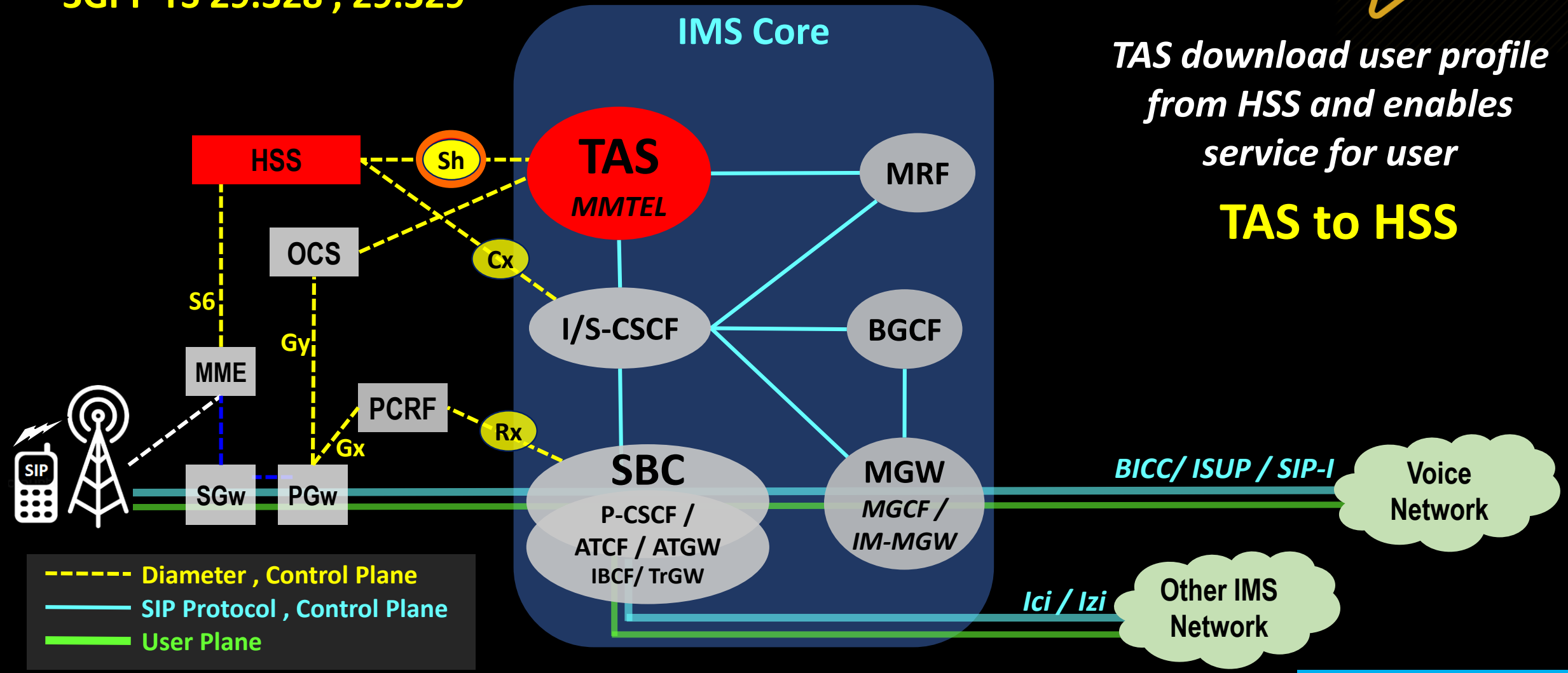
Sh DIAMETER

3GPP TS 29.328 , 29.329



WHY? Sh

TAS download user profile from HSS and enables service for user

TAS to HSS

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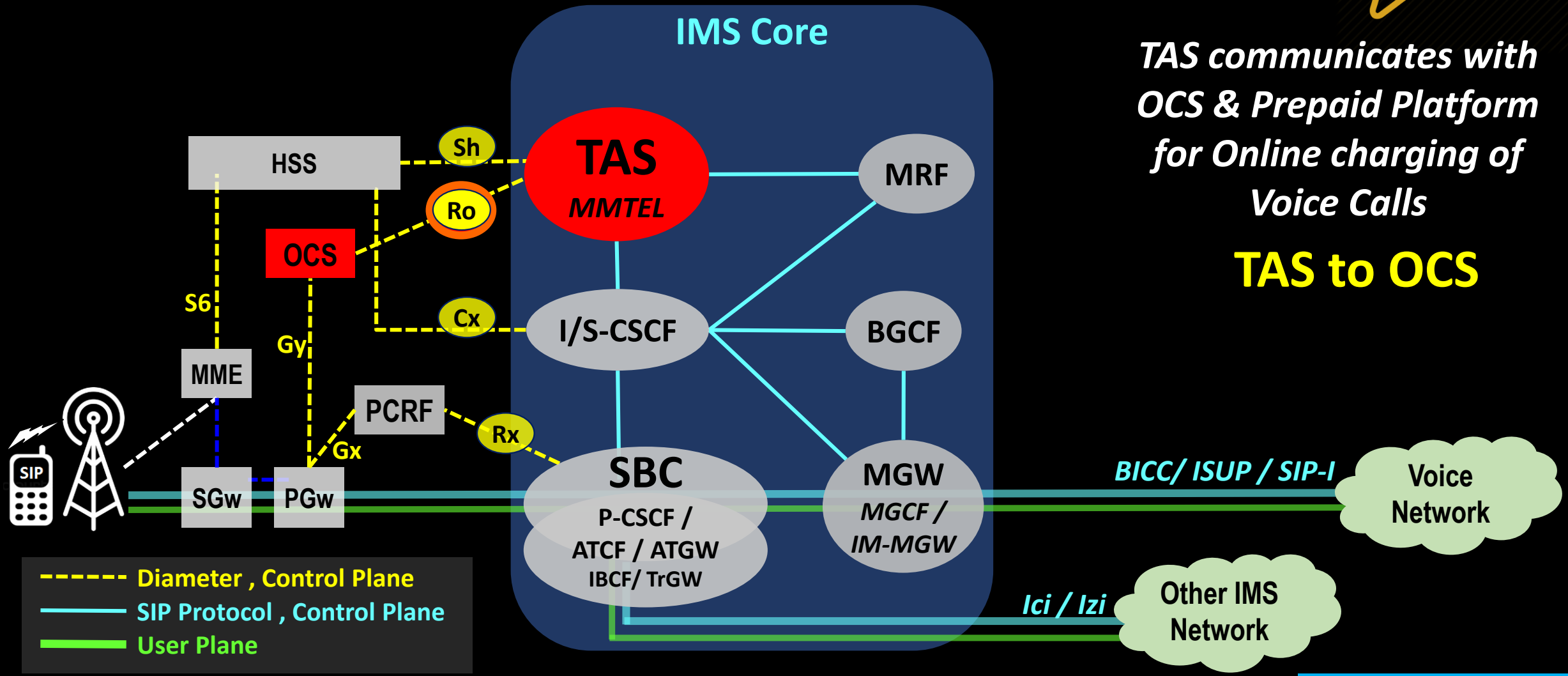
Ro DIAMETER

TAS communicates with OCS & Prepaid Platform for Online charging of Voice Calls

TAS to OCS

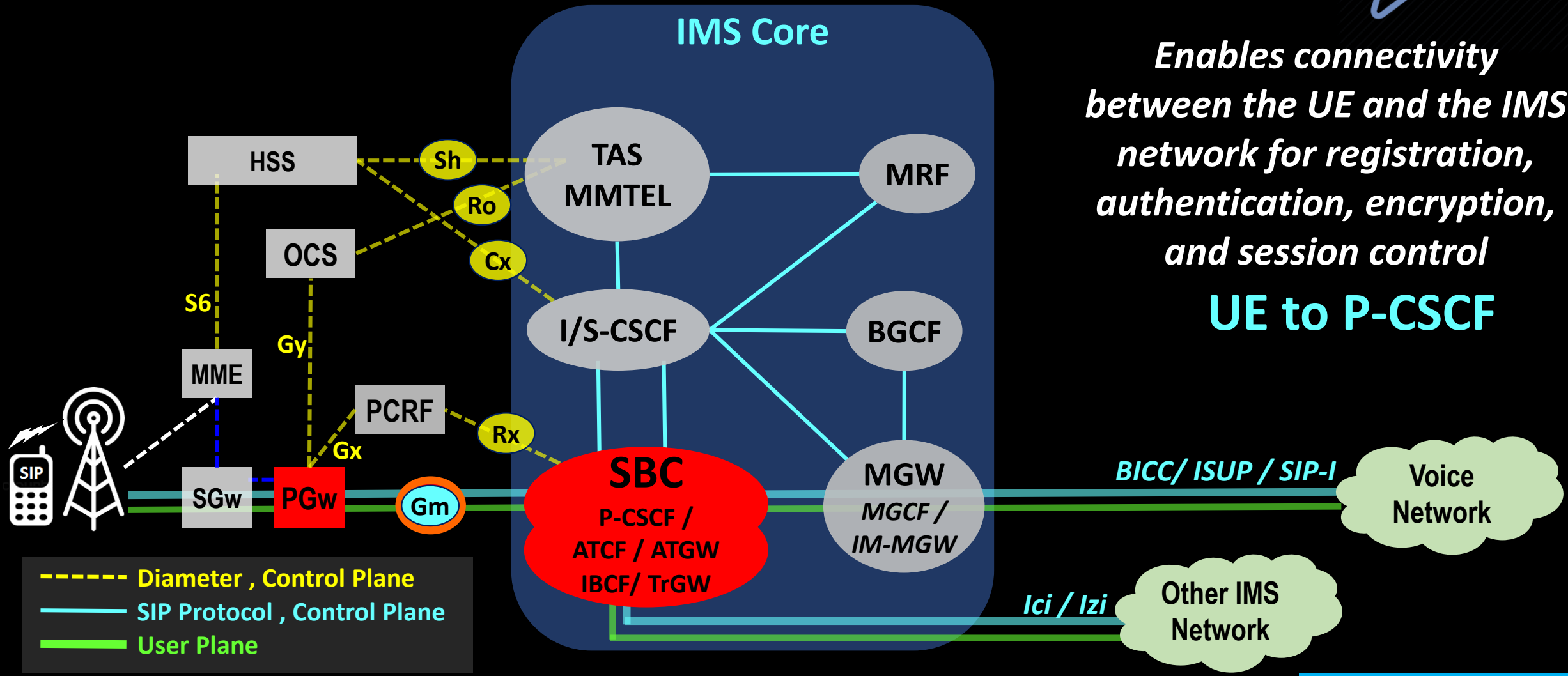


Gm

SIP / SDP

Enables connectivity between the UE and the IMS network for registration, authentication, encryption, and session control

UE to P-CSCF



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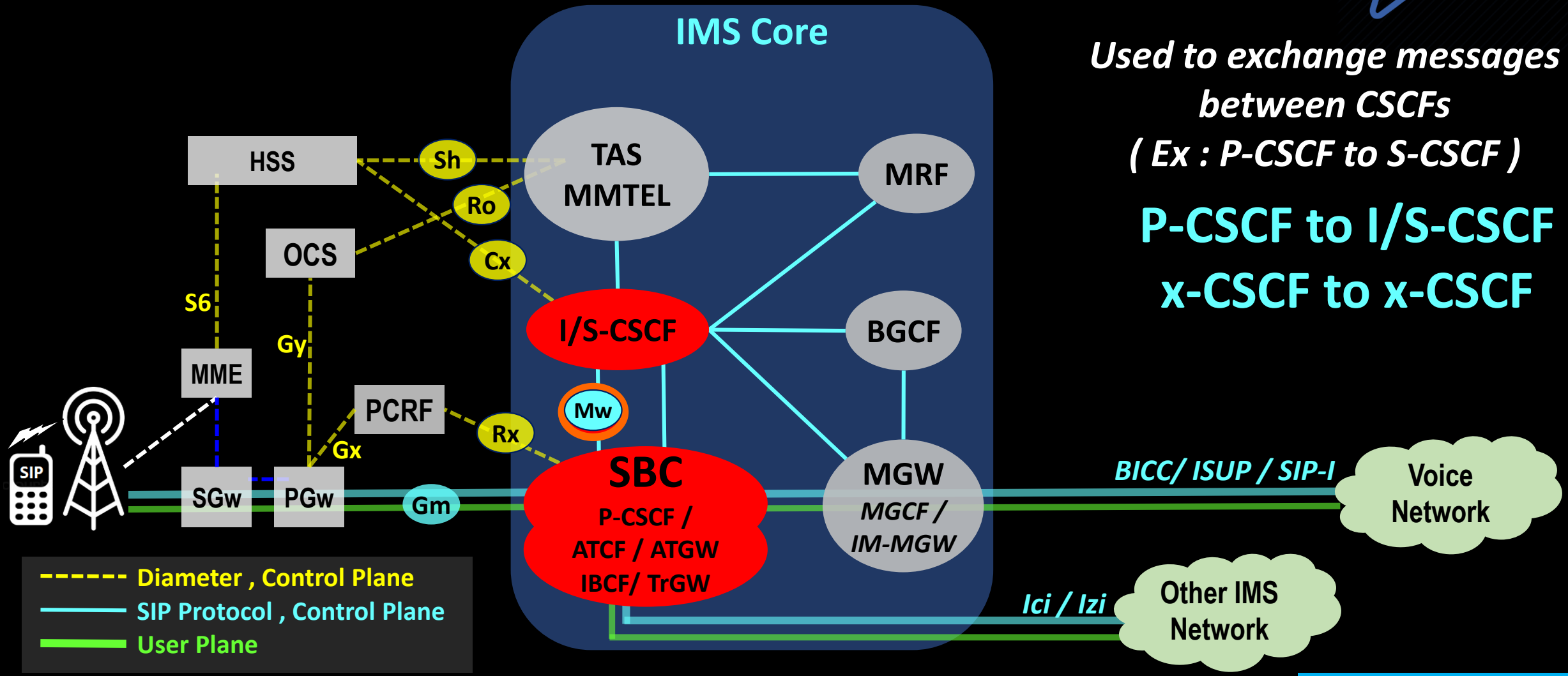
Mw

SIP / SDP

Used to exchange messages between CSCFs (Ex : P-CSCF to S-CSCF)

P-CSCF to I/S-CSCF
x-CSCF to x-CSCF



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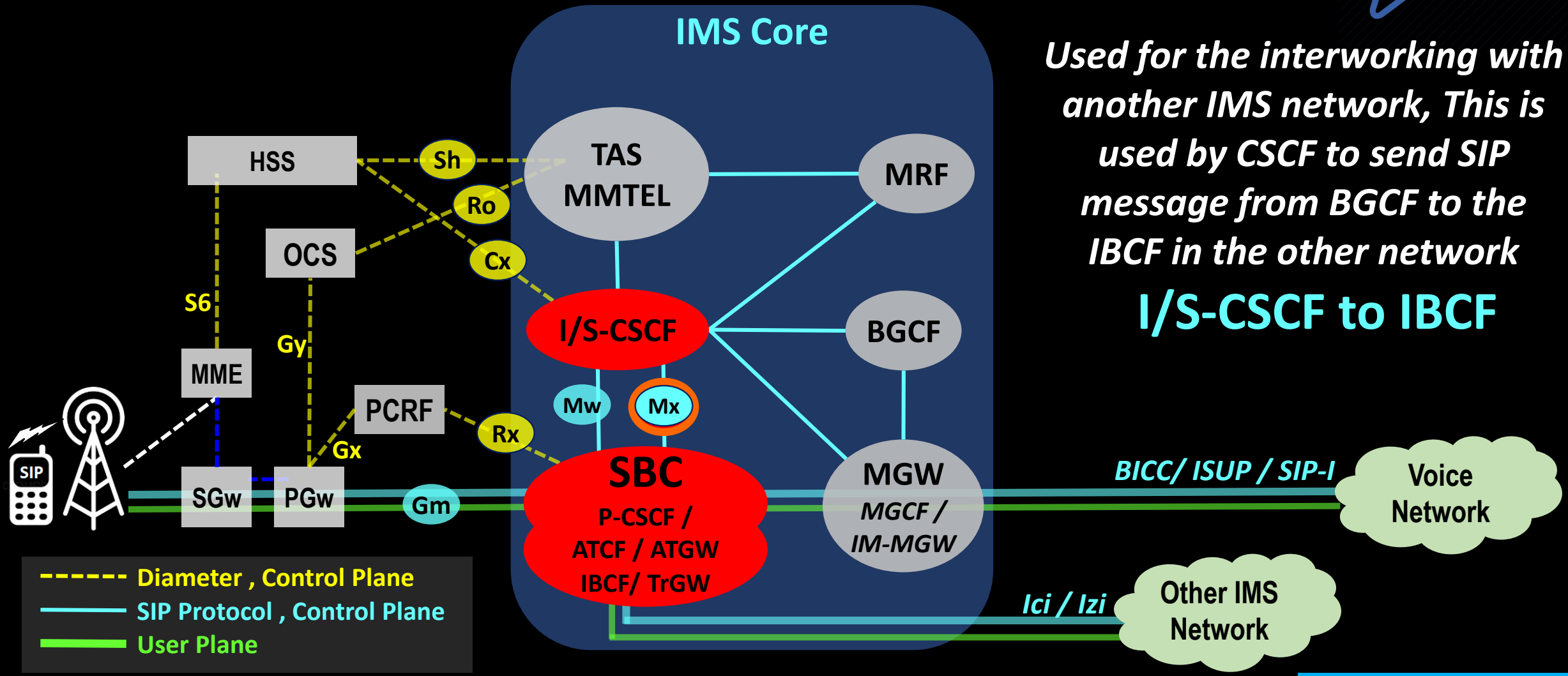
Mx

SIP / SDP

Used for the interworking with another IMS network, This is used by CSCF to send SIP message from BGCF to the IBCF in the other network

I/S-CSCF to IBCF



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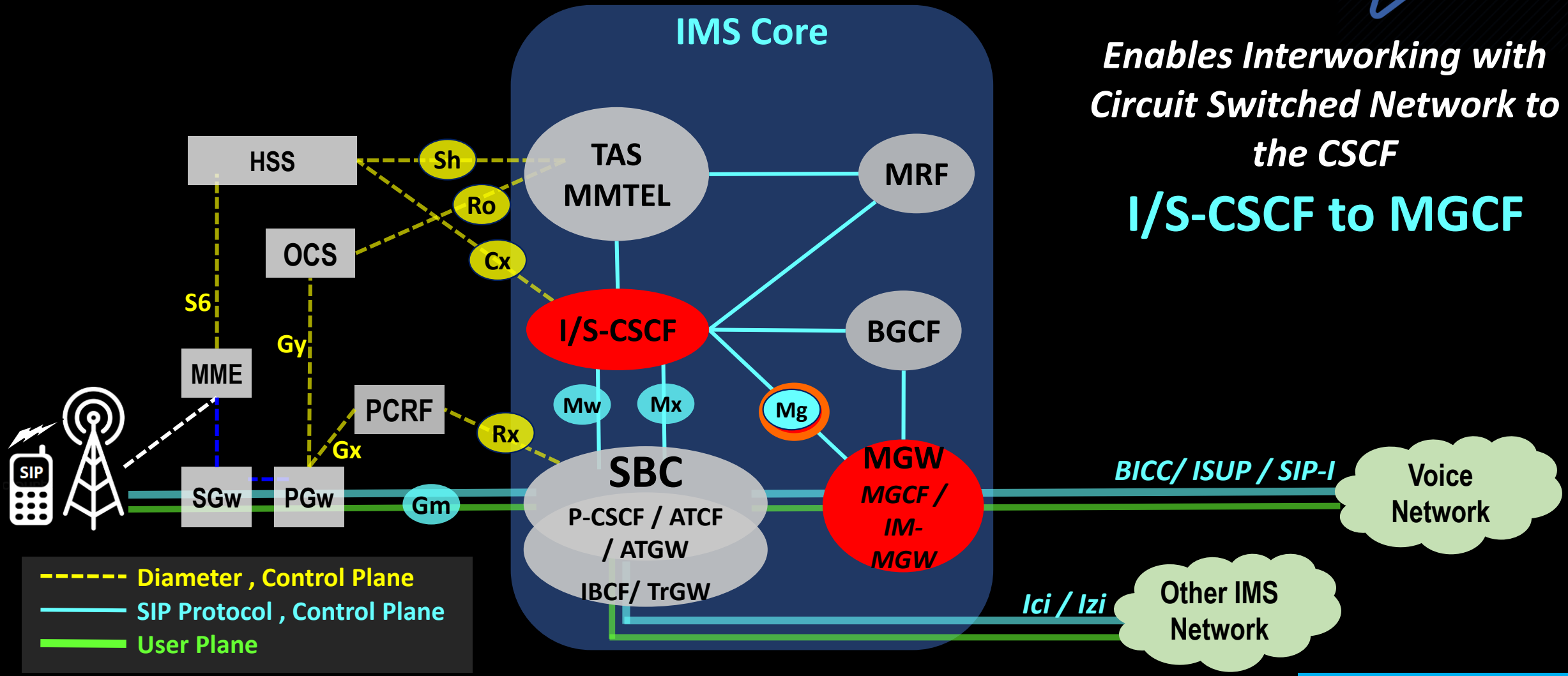
Mg

SIP / SDP

Enables Interworking with Circuit Switched Network to the CSCF

I/S-CSCF to MGCF



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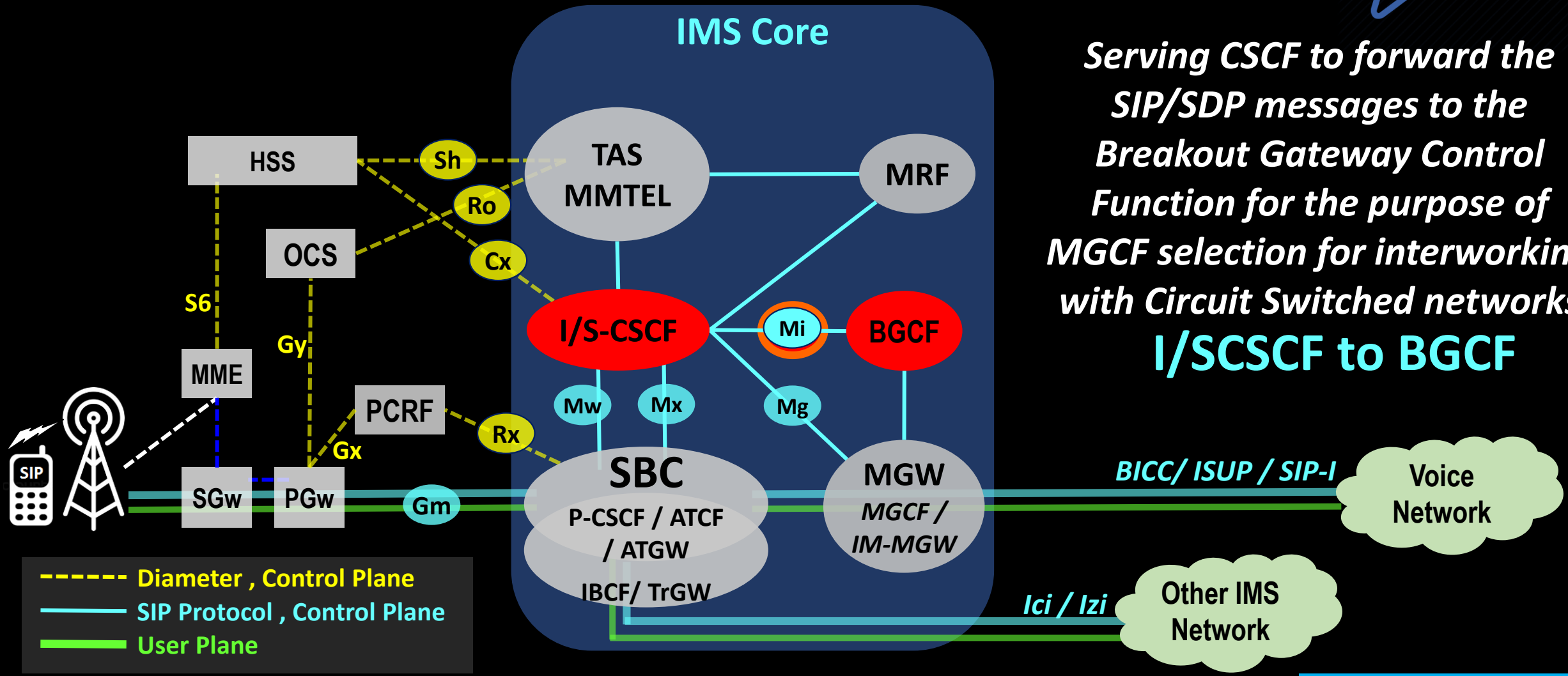
Mi

SIP / SDP

Serving CSCF to forward the SIP/SDP messages to the Breakout Gateway Control Function for the purpose of MGCF selection for interworking with Circuit Switched networks

I/SCSCF to BGCF



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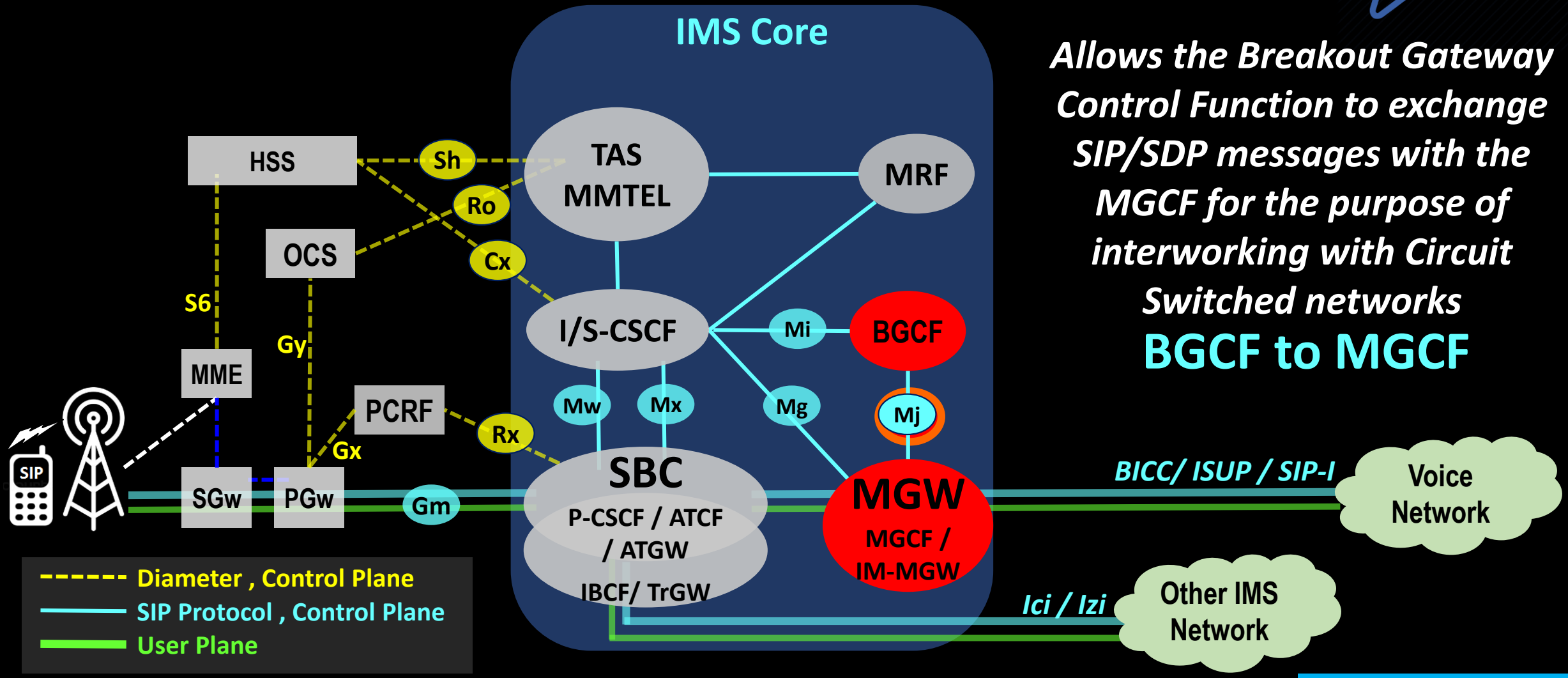
Mj

SIP / SDP

Allows the Breakout Gateway Control Function to exchange SIP/SDP messages with the MGCF for the purpose of interworking with Circuit Switched networks

BGCF to MGCF



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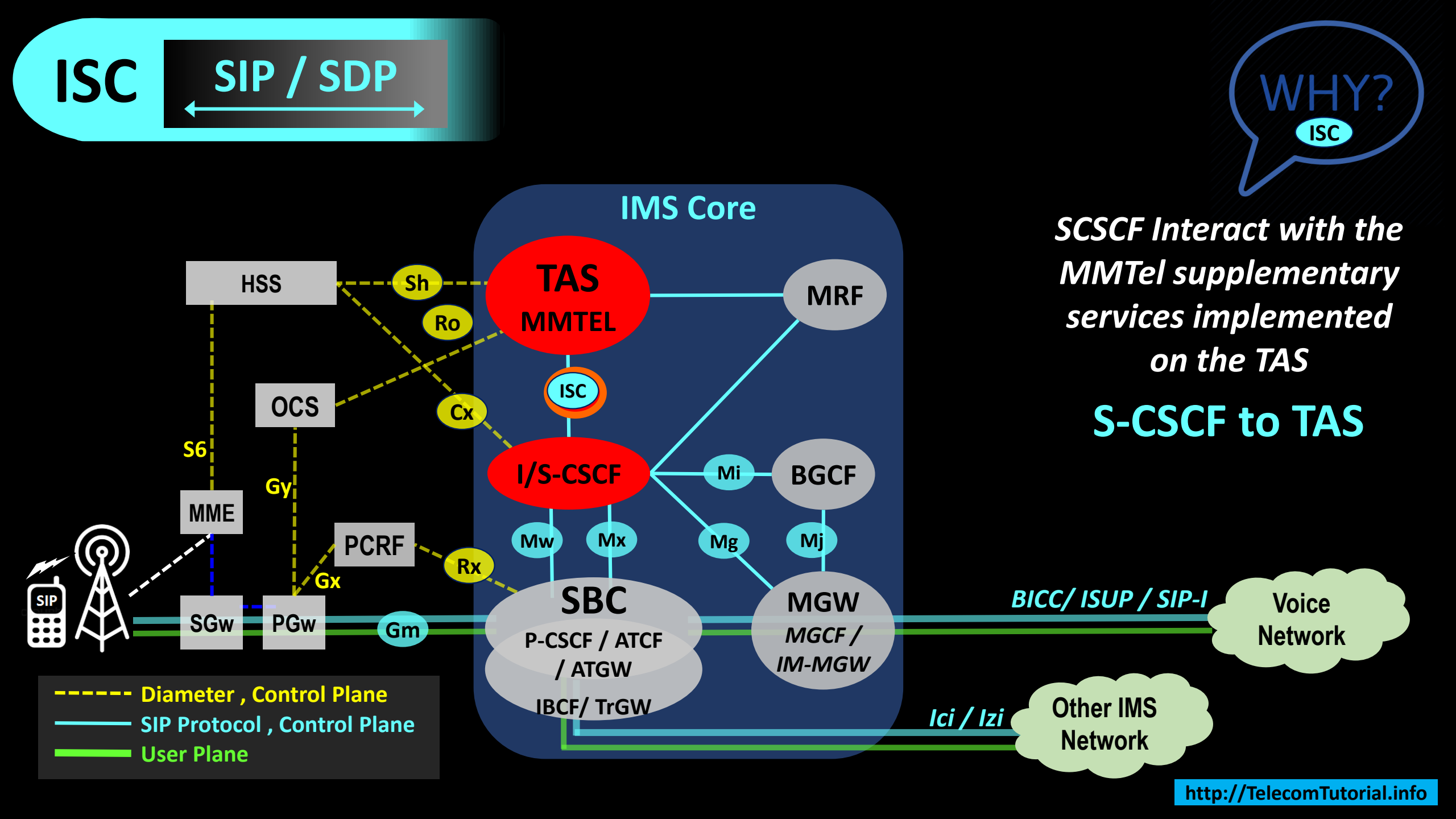
ISC
SIP / SDP
WHY?
ISC
SCSCF Interact with the MMTel supplementary services implemented on the TAS
S-CSCF to TAS
IMS Core
HSS
Sh
Ro
TAS
MMTEL
MRF
ISC
OCS
Cx
S6
Gy
I/S-CSCF
Mi
BGCF
MME
PCRF
Rx
Mw
Mx
Mg
Mj
Gx
SIP
SGw
PGw
Gm
SBC
P-CSCF / ATCF / ATGW
IBCF/ TrGW
MGW
MGCF / IM-MGW
BICC/ ISUP / SIP-I
Voice Network
Ici / Izi
Other IMS Network
Diameter , Control Plane
SIP Protocol , Control Plane
User Plane
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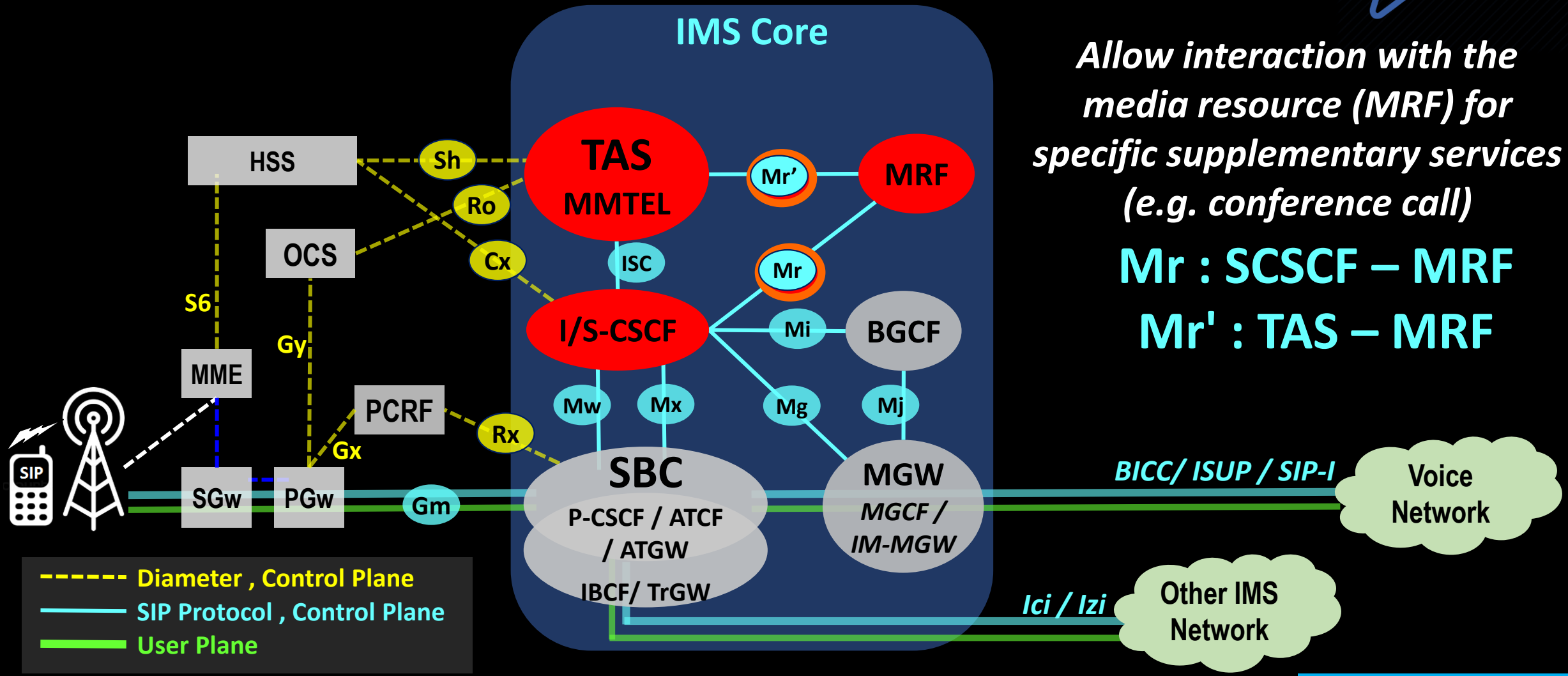
Mr & Mr’

SIP / SDP

Allow interaction with the media resource (MRF) for specific supplementary services (e.g. conference call)

Mr : SCSCF – MRF

Mr' : TAS – MRF



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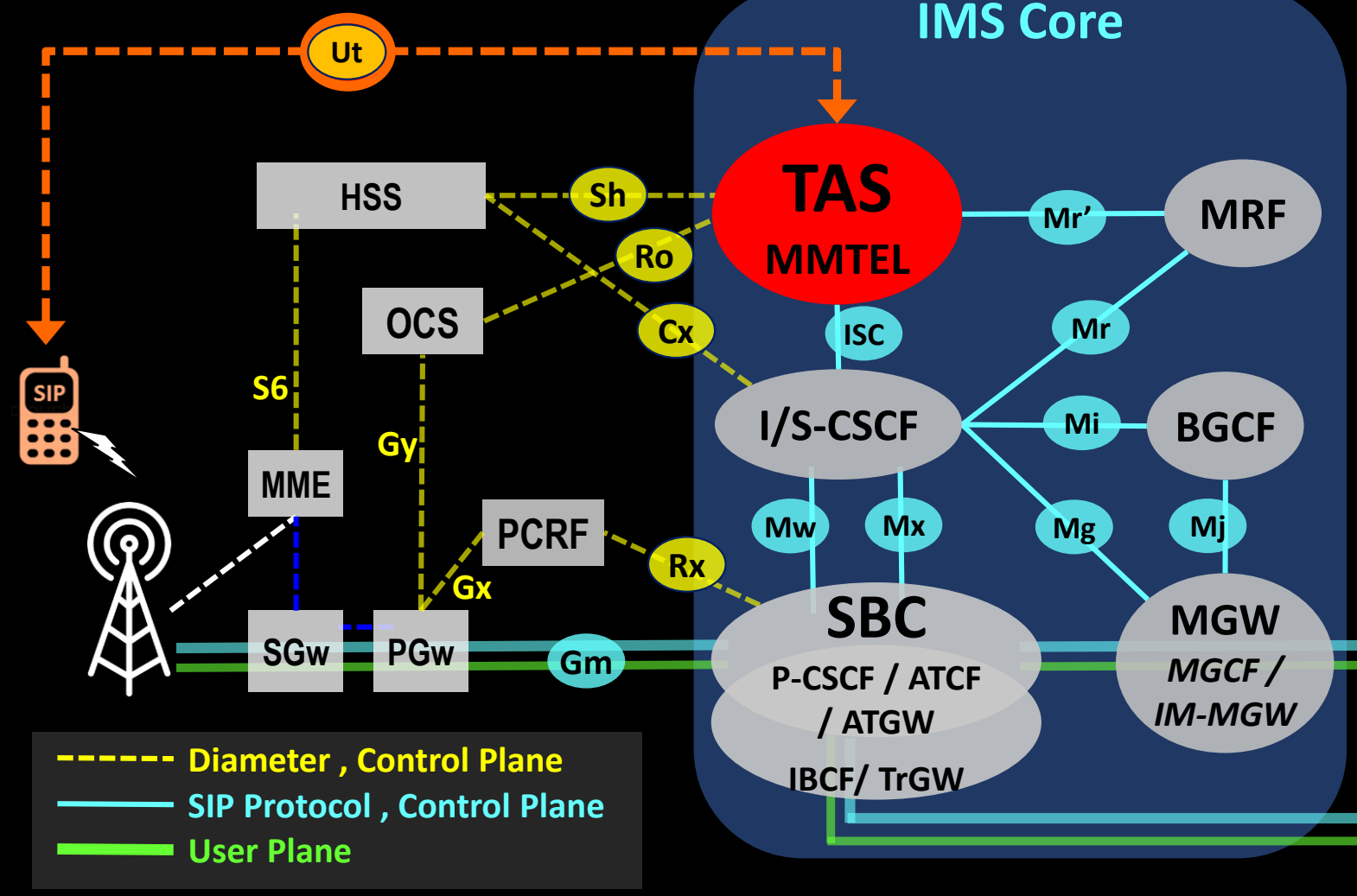
Ut XCAP

3GPP TS 24.623



Allows user configuration of the supplementary services specified for VoLTE service to Handset Directly

UE – TAS

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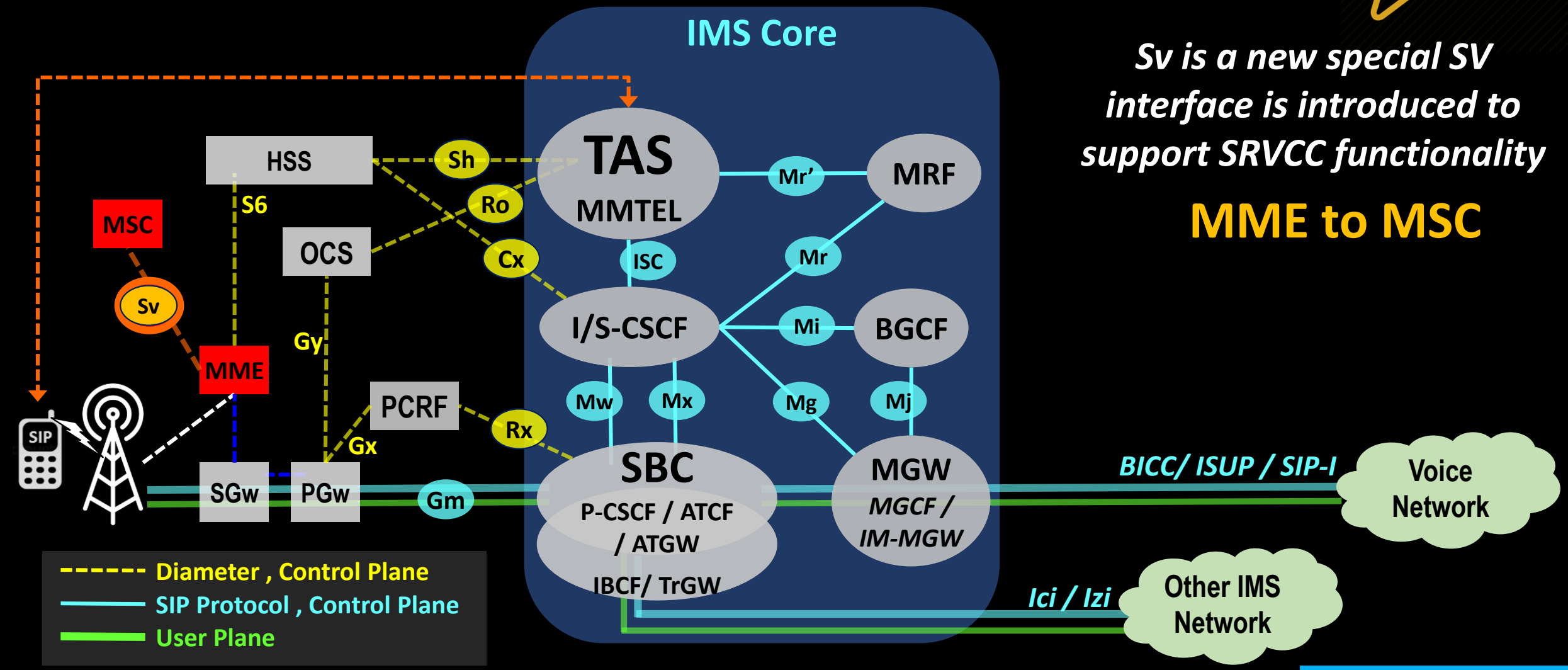
Sv GTPv2



Sv is a new special SV interface is introduced to support SRVCC functionality

MME to MSC

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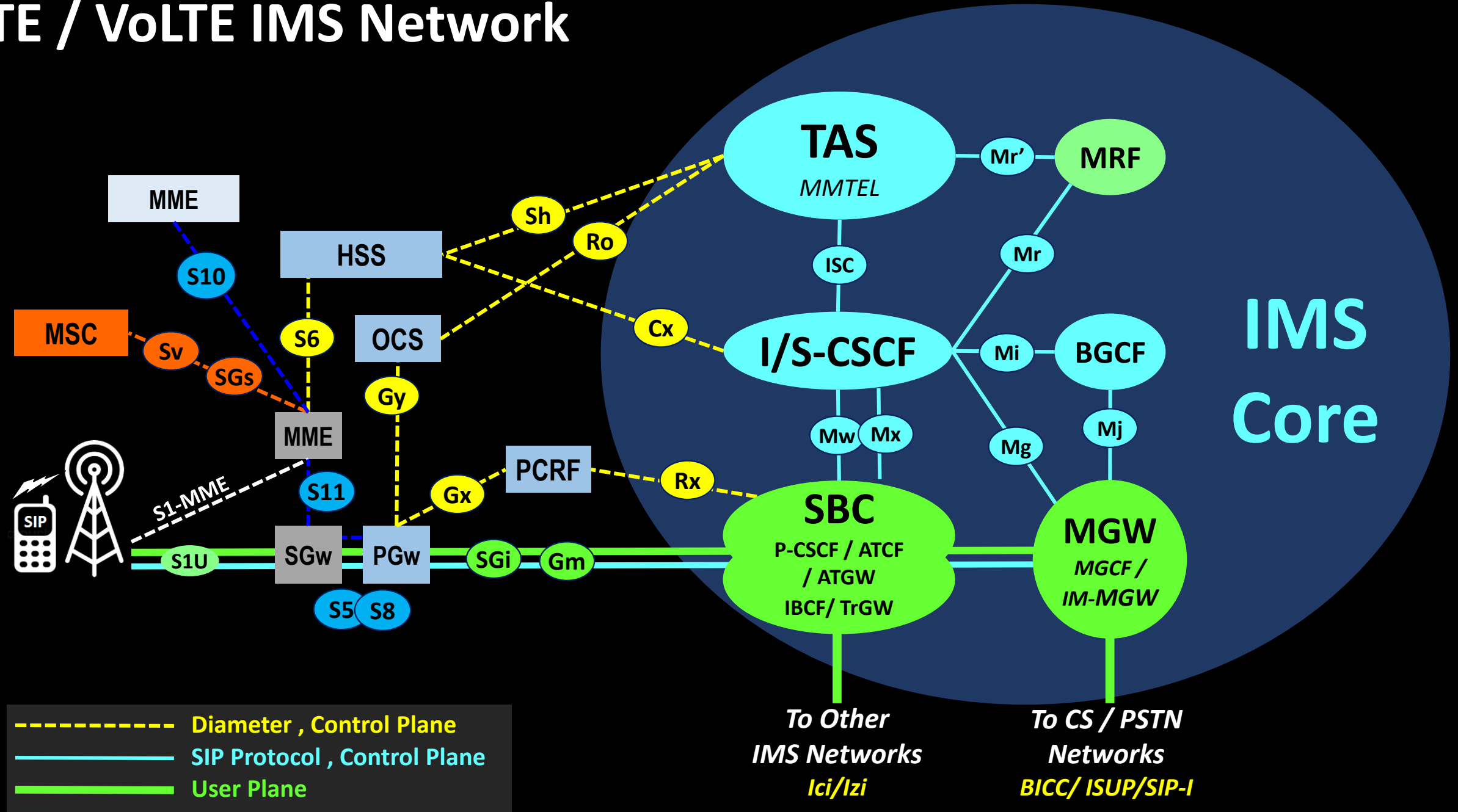
LTE / VoLTE IMS Network



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